

RAWDATA
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Metagenome Amplicon Sequencing **Report**

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Order Information

Client Name	이은수
Client Organization	건국대학교 글로벌캠퍼스
Order Number	HN00274756
Application	Metagenome Amplicon Sequencing
Type of Read	Paired-end
Read Length	301
Library Kit	Metagenome Amplicon
Type of Sequencer	illumina system

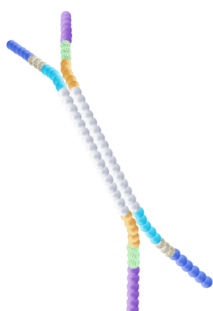
Experimental Workflow

The samples are prepared according to NGS library preparation workflow, and sequenced using Illumina platform. The workflow illustrated below shows the common ligation based method of library preparation. The process may differ based on the library preparation protocol followed.



Sample Preparation

DNA/RNA is first extracted from the sample, and samples which meet quality control standards proceed to library construction.



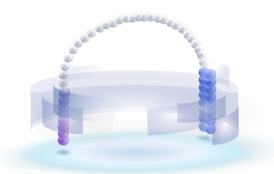
Ligate Adapters

The sequencing library is prepared by random fragmentation of the DNA or cDNA sample, followed by 5' and 3' adapter ligation. Alternatively, "tagmentation" combines the fragmentation and ligation reactions into a single step which greatly increases the efficiency of the library preparation process.

Final library Construction

Adapter-ligated fragments are then PCR amplified with a PCR primer solution which anneals to the ends of each adapters.

The library templates undergo quality control and quantification process.

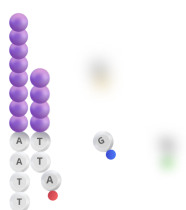


Cluster generation using bridge amplification

The library is loaded onto a flow cell where fragments are captured on a lawn of surface-bound oligos complementary to the library adapters.

Each fragment is then amplified into distinct clonal clusters through bridge amplification.

Once cluster generation is complete, the templates are ready for sequencing.



Sequencing by synthesis (SBS) technology

Illumina SBS technology utilizes a proprietary reversible terminator-based method that detects single bases as they are incorporated into DNA template strands. As all 4-reversible, terminator-bound dNTPs are present during each sequencing cycle, natural competition minimizes incorporation bias and greatly reduces raw error rates compared to other technologies.

The result is highly accurate base-by-base sequencing that virtually eliminates sequence-context-specific errors, even within repetitive sequence regions and homopolymers.



Generation of Raw data

The Illumina sequencer generates raw images utilizing sequencing control software for system control and base calling, through integrated primary analysis software called RTA (Real Time Analysis).

The BCL/cBCL (base call) binary files are converted into FASTQ files using bcl-convert which is an Illumina provided package. Adapters are not trimmed away from the reads.

Raw Data Statistics

The total number of bases, reads, GC (%), Q20 (%), and Q30 (%) are calculated for the 10 sample(s).
 For example, in 10_P_densiflora sample, 167,986 reads are produced, and total read bases are 50.6 Mbp.
 The GC content (%) is 48.6% and Q30 is 96.0%.

* Raw Data

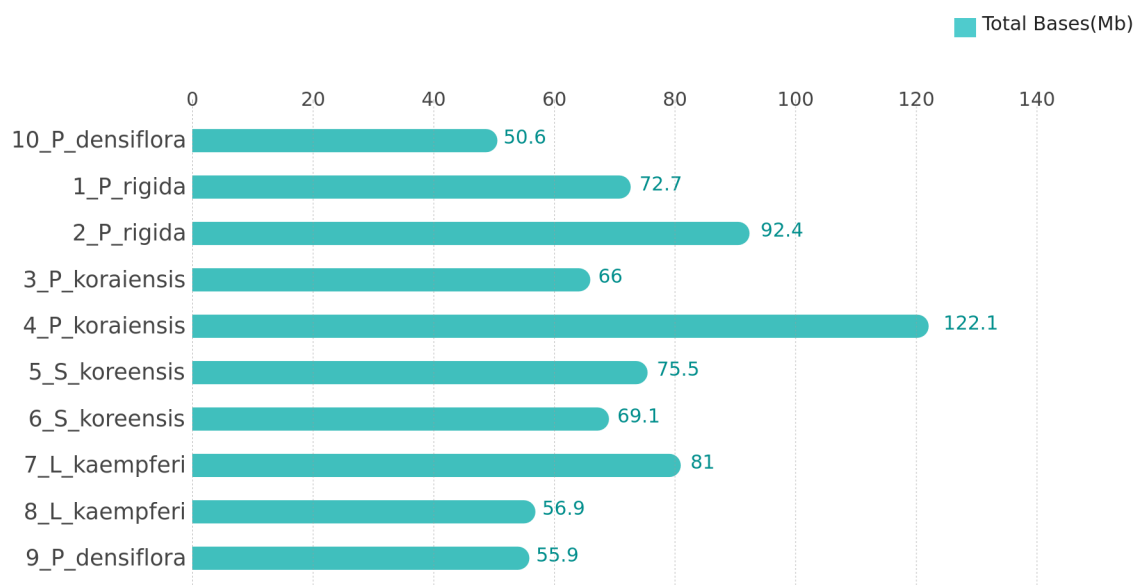
Sample ID	Total bases (bp)	Total reads	GC (%)	AT (%)	Q20 (%)	Q30 (%)
10_P_densiflora	50,563,786	167,986	48.6	51.4	98.7	96.0
1_P_rigida	72,724,008	241,608	53.4	46.6	98.7	96.1
2_P_rigida	92,402,184	306,984	55.6	44.4	98.7	96.0
3_P_koraiensis	66,018,932	219,332	51.3	48.7	98.6	95.9
4_P_koraiensis	122,140,382	405,782	55.8	44.2	98.8	96.2
5_S_koreensis	75,481,168	250,768	52.0	48.0	98.7	96.0
6_S_koreensis	69,077,092	229,492	51.6	48.4	98.8	96.3
7_L_kaempferi	80,994,284	269,084	51.3	48.7	98.7	95.9
8_L_kaempferi	56,851,676	188,876	53.4	46.6	98.7	96.0
9_P_densiflora	55,857,172	185,572	52.8	47.2	98.8	96.2

- Sample ID : Sample name.
- Total Bases (bp) : Total number of bases sequenced.
- Total Reads : Total number of reads. For illumina paired-end sequencing, this value refers to the sum of read1 and read2.
- GC (%) : Ratio of GC content.
- AT (%) : Ratio of AT content.
- Q20 (%) : Ratio of bases that have phred quality score of over 20.
- Q30 (%) : Ratio of bases that have phred quality score of over 30.

Total Bases

Total number of samples : 10

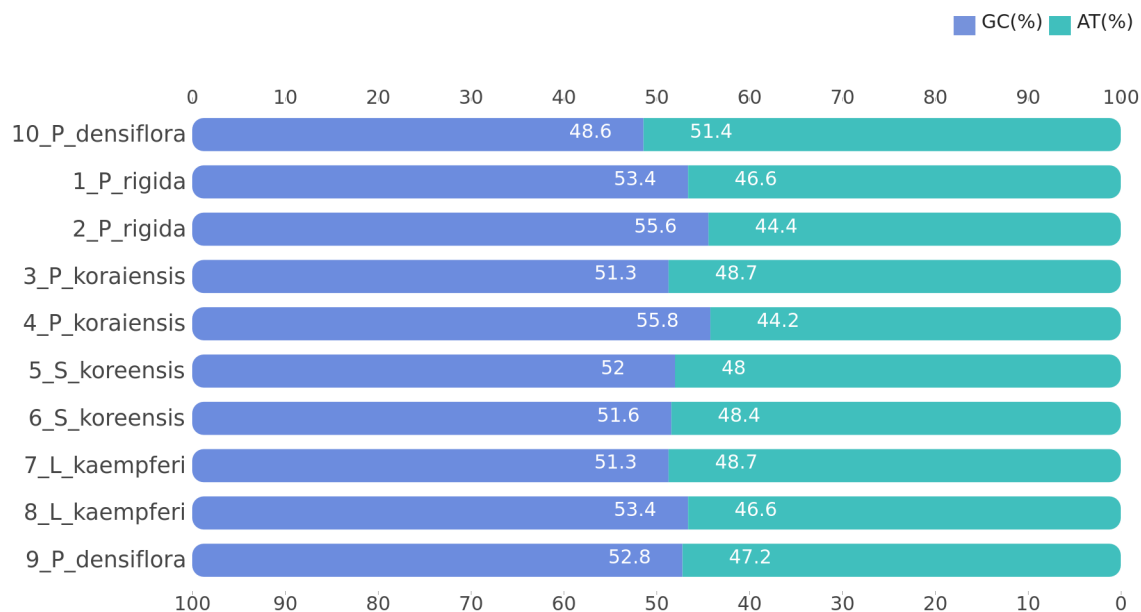
* Raw Data



GC/AT Content

Total number of samples : 10

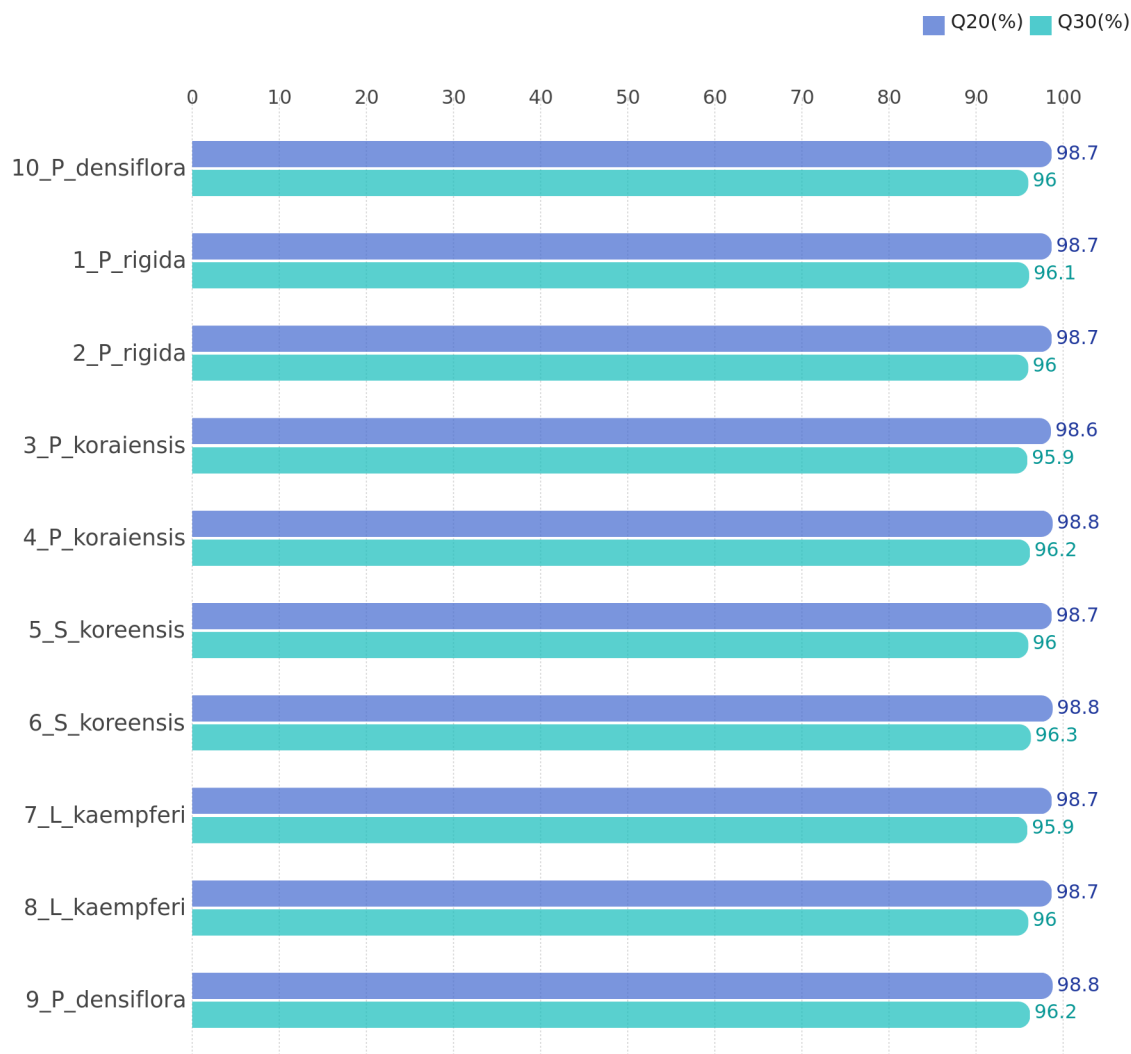
* Raw Data



Q20/Q30 (%)

Total number of samples : 10

* Raw Data



Download List

- The following links provide access to your data files. Please download all files within 2 weeks, as the links will expire thereafter.
- Once the download is complete, kindly check the file integrity (e.g., file size and checksum) before proceeding with analysis. If you encounter any issues with the download, please contact us for assistance.

* Raw Data Download

File Name	File Size(byte)	md5sum
HN00274756.zip	72,859,791	53f8ec0893c84286072a3bbf1030ac22

File Name	File Size(byte)	md5sum
4_P_koraiensis_1.fastq.gz	6,003,740	6a3c053413aee740be1587da9b09ce33
4_P_koraiensis_2.fastq.gz	6,079,889	b4da09f064dd00f270d88b39c9dfa4ba
5_S_koreensis_1.fastq.gz	3,624,253	308aa926d01875812a87aea2848be2a0
5_S_koreensis_2.fastq.gz	3,723,808	fbfc83e3e1db7c858e6947533598e234
7_L_kaempferi_1.fastq.gz	3,802,601	ae9162fe49c8e5620a19521e971a0a5d
7_L_kaempferi_2.fastq.gz	3,911,291	9dd7460f414fcac7179d26c6d7baaf28
8_L_kaempferi_1.fastq.gz	2,788,393	dd54c8e8aef597464666fe9d588c1e08
8_L_kaempferi_2.fastq.gz	2,811,747	ab5657466c93016594bcee6c2be91baf
6_S_koreensis_1.fastq.gz	3,247,413	c61cda2176a24d6bb63d319e362995d7
6_S_koreensis_2.fastq.gz	3,258,764	7e14cf778966f295ddecffcbde2aea2e
1_P_rigida_1.fastq.gz	3,682,076	2c17e17a11597d3b86818fb8da302cd9
1_P_rigida_2.fastq.gz	3,755,224	6f75e056d2df19679ee35ad97a770b4b
3_P_koraiensis_1.fastq.gz	3,119,652	1e51c3f9133d4dd46c6149aa0b1caf20
3_P_koraiensis_2.fastq.gz	3,073,310	a7af2b65e0f3c05d79cd16b63e854741
9_P_densiflora_1.fastq.gz	2,738,744	167d53e8f51870ad05f30d1b12ab12b1
9_P_densiflora_2.fastq.gz	2,773,574	b04e35ab2106c61130eb4cae1c072bf4
2_P_rigida_1.fastq.gz	4,789,966	2ddb23b23ae9bb1e648595ea1776837a
2_P_rigida_2.fastq.gz	4,769,052	db036e46f9c8f485efa0deea85d758e5
10_P_densiflora_1.fastq.gz	2,710,689	a04b1242a2ce1044cac9e778c72adf9a
10_P_densiflora_2.fastq.gz	2,192,067	7df0b15a7dcf1e2a4a77a4d914da83ed

FAQ

Q Why do I need to check the md5sum values, and how can I check it? (Windows system)

A NGS data tend to have a large files size which makes them more likely to be corrupted during file transfer. So it's important that you check the md5sum of the files after receiving them to make sure what you received are what we gave.

Checking md5 hash in a Windows system

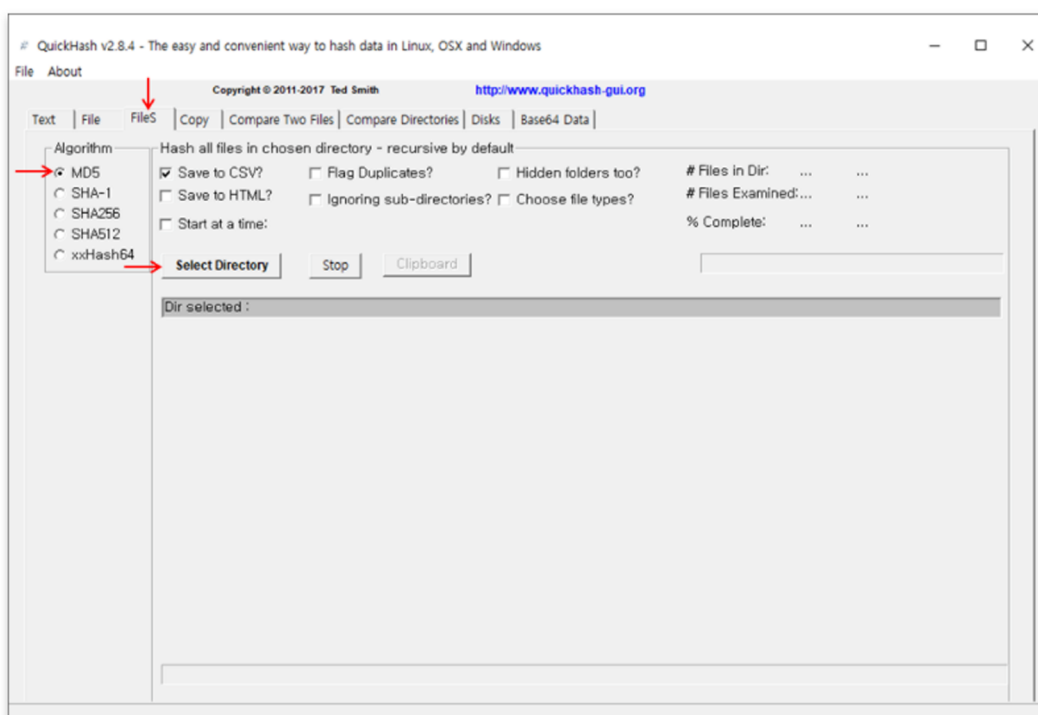
Windows does not provide a program for checking md5sum by default. An external program such as [QuickHash-Windows](#) can be used instead.

STEP 1 Download QuickHash-Windows from the website, and unzip the file.

STEP 2 Take a look at the UserManual.pdf file inside the zip file, and execute the .exe file.

Quickhash-GUI.exe	2,090,414	6,505,472
sqlite3-win32.dll	429,646	852,754
sqlite3-win64.dll	717,149	1,742,848
UserManual.pdf	512,697	576,987

STEP 3 Click on the "FileS" tab, and select [MD5] as the Algorithm.



STEP 4 Click "Select Directory" and choose the directory where the files to be checked are located in. The output can be saved as a csv or txt file. The process may take some time depending on the performance of the system being used.

STEP 5 Compare the newly calculated md5 value with the md5 value provided to you through the Analysis Report.

FAQ

Q Why do I need to check the md5sum values, and how can I check it? (Linux system)

A NGS data tend to have a large files size which makes them more likely to be corrupted during file transfer. So it's important that you check the md5sum of the files after receiving them to make sure what you received are what we gave.

Checking md5 hash in a Linux system

Linux systems have an internal md5sum program under /user/bin/md5sum.

md5sum has a "-c" option, which reads the MD5 sums from the input file and checks them simultaneously.

Usage: \$ md5sum -c [input file name]

STEP 1 Macrogen provides a text file containing the md5sum of deliverables you'll be receiving, which you can use to validate the integrity of the files. You can download this file by clicking on the "md5sum List" button in the "Download List" page. The text file will have the following name and format depending on how you're receiving your data:

o Via download link : <OrderNumber>_#samples_md5sum_DownloadLink.txt

```
[user@host] cat HN00000000_1samples_md5sum_DownloadLink.txt
File      Size      md5sum      Download link
test_1.fastq.gz 3118212249  07a66a1d7d7fde2ee71b02a2caf21aba  https://data.macrogen.com/-macro3/HiSeq02/20210322/HN00000000/test_1.fastq.gz
test_2.fastq.gz 3305438294  3b4ff911e5d238a3c4763ee7967cb29a  https://data.macrogen.com/-macro3/HiSeq02/20210322/HN00000000/test_2.fastq.gz
```

o Via HDD : <OrderNumber>_#samples_md5sum.txt

```
[user@host] cat HN00000000_1samples_md5sum.txt
File      Size      md5sum
test_1.fastq.gz 3118212249  07a66a1d7d7fde2ee71b02a2caf21aba
test_2.fastq.gz 3305438294  3b4ff911e5d238a3c4763ee7967cb29a
```

o You can also find "md5sum.txt" located inside the HDD delivered to you.

```
[user@host]$ cat md5sum.txt
07a66a1d7d7fde2ee71b02a2caf21aba  RawData/test_1.fastq.gz
3b4ff911e5d238a3c4763ee7967cb29a  RawData/test_2.fastq.gz
```

STEP 2 Use "md5sum -c" to validate the integrity of the file you've received. The input file for md5sum -c has to be delimited by two spaces with the md5sum column appearing before the file name, just like the sample image of "md5sum.txt" file shown above. As you can see, the two other files above are not formatted this way and need to be altered to be used as input for md5sum -c. You can manually exclude the header and cut out "File" and "md5sum" column from the files, or simply run the following command:

\$ awk '{print \$3 " " " \$1}' <md5sum_file> | grep -v File

STEP 3 "md5sum -c" reads the input containing the md5 value of a file, and checks whether the md5 value of that file matches what's written inside the input file. This action outputs "OK" if the md5 value matches, and "FAILED" if otherwise. Check if the command outputs "OK" for all the files. (Refer to image below)

```
user@host
[user@host] awk '{print $3 " " " $1}' HN00000000_1samples_md5sum_DownloadLink.txt | grep -v File > md5sum.txt
[user@host] cat md5sum.txt
07a66a1d7d7fde2ee71b02a2caf21aba  test_1.fastq.gz
3b4ff911e5d238a3c4763ee7967cb29a  test_2.fastq.gz
[user@host]
[user@host] md5sum -c md5sum.txt
test_1.fastq.gz: OK
test_2.fastq.gz: OK
[user@host]
```

FAQ

Q I want to see the data produced by MacroGen. How can I open the files?

A NGS data tend to have large file sizes, and are not user-friendly to work with in a Windows environment. We recommend that you use Linux system for smoother operation.

Q Where can I find information for Illumina adapter sequences?

A Information on Illumina adapters can be found in this support document:
[Adapter Sequences Intro](#)

Result File Description

Deliverables List

File Type	File Name	Description
FASTQ	 [Sample name]_[read1].fastq.gz	Raw read1 sequence data
	 [Sample name]_[read2].fastq.gz	Raw read2 sequence data
md5sum	[Order#]_[#samples]_md5sum[DownloadLink].txt	You can download this file by clicking on the "md5sum List" button found on the "Download List" page. The file is slightly different in terms content, depending on how you're receiving your data. If you're receiving via download link, the file contains the following information : File name, File size, md5sum, FTP link. Otherwise, if your receiving your data via HDD the file only contains : File name, File size, and md5sum. MD5 is a string of 32 hexadecimal values, which represents a 'fingerprint' of a file. By comparing the supplied MD5 value to the actual value computed by the MD5sums utility, you can make sure that the file that you downloaded off of the internet has not been tampered with or modified from the original file stored in our server.

FASTQ Format

Example:

Line 1 : Sequence identifier

Line 2 : Nucleotide sequences

Line 3 : Quality score identifier line - character '+'

Line 4 : Quality score

```

@A00125:17:H2HFJDMXX:1:1101:3170:1000 1:N:0:ATGCCTAA
GAAACACGATGACACTCACATGGCACTCACATTTTCAGCTCCTTTCTAAGTGATTGCAAATATTAATTCATAT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00125:17:H2HFJDMXX:1:1101:9408:1000 1:N:0:ATGCCTAA
TGTGCGAAGGAAATCATTTCAGATGACAGTGTTAACCATGGTCAAAGGACCATTCTGTCTATCCTTCTTA
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF

```

FASTQ file consists of four lines.

Quality score is represented with each character.
One character matches its base with Phred+33

Phred Quality Score

Phred quality score numerically expresses the accuracy of each nucleotide. Higher Q number signifies higher accuracy. For example, if Phred assigns a quality score of 30 to a base, the chances of having base call error are 1 in 1000. Phred Quality Score Q is calculated with $-10\log_{10}(P)$, where P is probability of erroneous base call.

Quality of phred score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10000	99.99%



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